INCH-POUND

MIL-DTL-17/112B
19 June 2015
SUPERSEDING
MIL-C-17/112A
27 April 1981

MILITARY SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, FLEXIBLE COAXIAL, 50 OHMS, M17/112-RG304

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

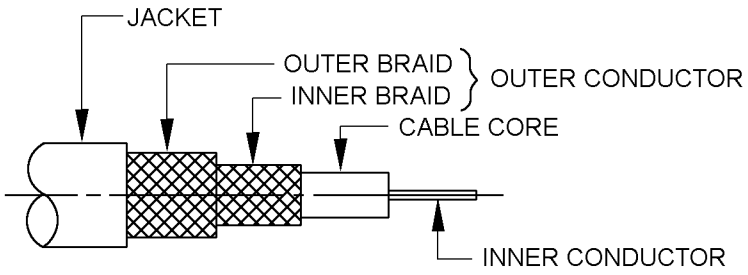


FIGURE 1. Configuration.

TABLE I. Description.

Components	Construction details
Inner conductor	Silver covered, copper-covered steel wire. Overall diameter: .059 inch ± .001.
Cable core	Type F-1: Overall diameter: .185 inch ± .005.
Outer conductor Inner braid Outer braid	Double braid, of AWG size 34 silver-covered wire. Overall diameter: .250 inch maximum. Carriers: 24 Ends: 5 Picks/inch: 14.5 ±10% Carriers: 24 Ends: 6 Picks/inch: 11.5 ±10%
Jacket	Type IX. Overall diameter: .280 inch ± .008. Minimum wall thickness: .015 inch.

AMSC N/A

FSC 6145

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:11Z
Check the source to verify that this is the current version before use.

ENGINEERING INFORMATION:

Continuous working voltage: 2,200 V rms, maximum.
Operating frequency: 12.4 GHz, maximum.
Velocity of propagation: 69.5 percent, nominal.
Power rating: See figure 2.
Operating temperature range: -55°C to +200°C.
Inner conductor properties:
DC resistance (maximum at 20°C): 0.8 ohm per 100 feet.
Elongation: 1 percent, minimum.
Tensile strength: 110 $klb_f/inch^2$, minimum.

Engineering notes: This cable is useful in general purpose, high temperature applications (see connector series “TNC” and “BNC” per MIL-PRF-39012).

REQUIREMENTS:

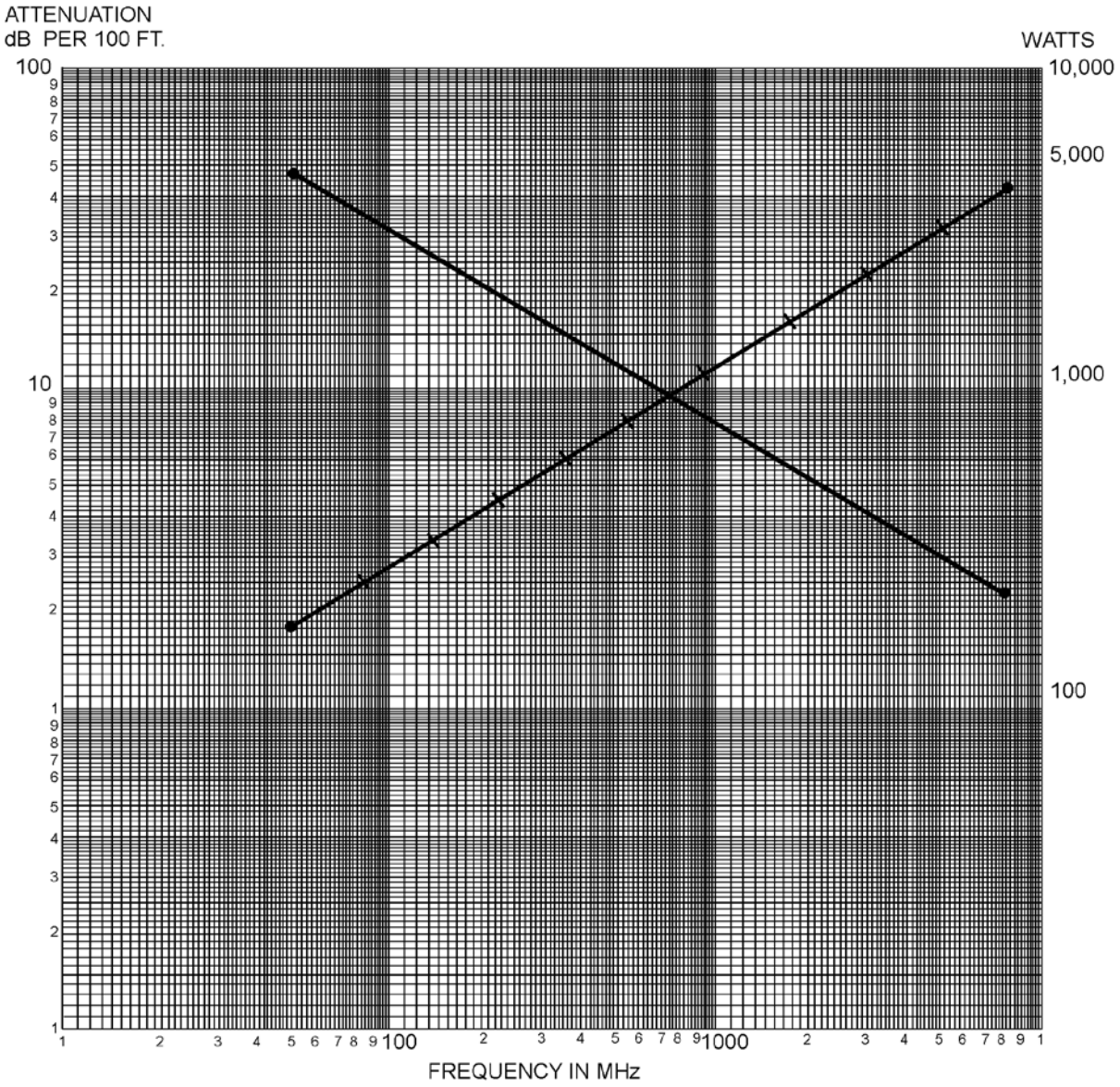
Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:
Adhesion of conductors:
Inner conductor to core: 4 pounds, minimum; 30 pounds, maximum.
Aging stability: Not applicable.
Stress crack resistance: +230° ± 5°C; mandrel size seven times the jacket diameter.
Outer conductor integrity: Not applicable.
Cold Bend: -55° ± 2°C.
Dimensional stability: +200°C ± 5°C.
Inner conductor from core: .250 inch, maximum.
Inner conductor from jacket: .312 inch, maximum.
Bendability: Not applicable.
Flammability: Applicable.
Weight: 9.4 pounds per 100 feet, maximum.

Electrical:

Spark test: 2,000 Vrms, +10 percent, -0 percent.
Voltage withstanding: 5,000 Vrms, +10 percent, -0 percent.
Insulation resistance: Not applicable.
Corona extinction voltage: 3,000 Vrms, minimum.
Characteristic impedance: 50 ± 2 ohms.
Attenuation: See figure 2.
Structural return loss: See figure 3.
Capacitance: 32 pF per foot, maximum.
Capacitance unbalance: Not applicable.
Transmission unbalance: Not applicable.
Mechanically induced noise voltage: Not applicable.
Time delay: Not applicable.
Contamination: Not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:11Z
Check the source to verify that this is the current version before use.



Tabulated values are for reference only. The values on the chart represent the requirement.

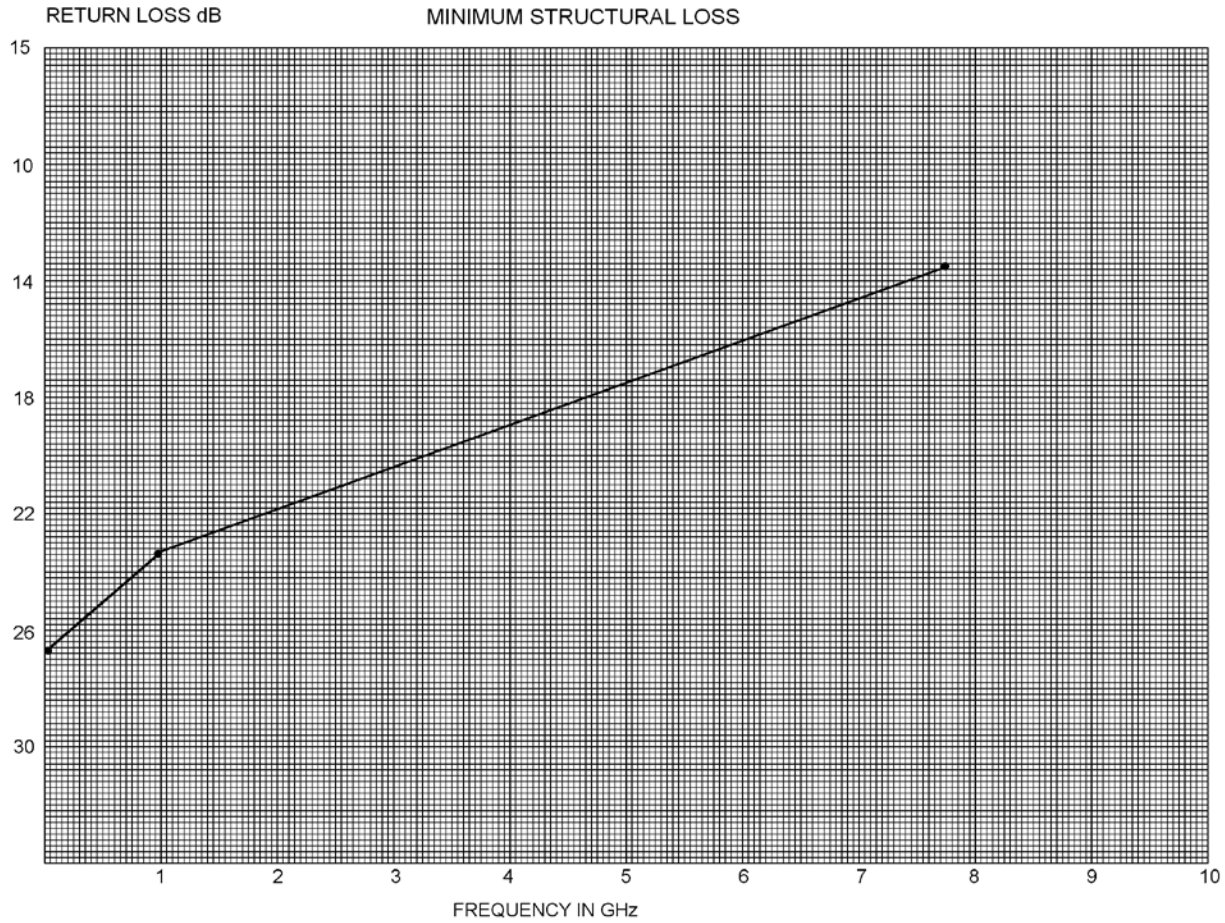
Maximum attenuation at 25°C sea level

Maximum power at 25°C, sea level

Frequency MHz	Atten dB	Power Watts
50	1.8	4600
10	2.7	3200
400	6.4	1450
1000	11.1	870
8000	40	210

FIGURE 2. Power rating and attenuation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:11Z
Check the source to verify that this is the current version before use.



SWR	REFLECTION COEFFICIENT	RETURN LOSS dB	SWR	REFLECTION COEFFICIENT	RETURN LOSS dB
17.3910	.8913	1	1.3767	.1585	16
8.7242	.7943	2	1.3290	.1413	17
5.8480	.7079	3	1.2880	.1259	18
4.4194	.6310	4	1.2528	.1122	19
3.5698	.5623	5	1.2222	.1000	20
3.0095	.5012	6	1.1957	.0891	21
2.6146	.4467	7	1.1726	.0794	22
2.3229	.3981	8	1.1524	.0708	23
2.0999	.3548	9	1.1347	.0631	24
1.9250	.3162	10	1.1192	.0562	25
1.7849	.2818	11	1.1055	.0501	26
1.6709	.2512	12	1.0935	.0447	27
1.5769	.2239	13	1.0829	.0398	28
1.4985	.1995	14	1.0736	.0355	29
1.4326	.1778	15	1.0653	.0316	30

MHz	dB
50	26.6
400	25.2
1000	23.2
8000	13.2

Tabulated values are for reference only. The values on the chart represent the requirements.

FIGURE 3. Structural return loss.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:11Z
Check the source to verify that this is the current version before use.

Part or Identifying Number (PIN): M17/112-RG-304.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Referenced documents. In addition to MIL-DTL-17, this document references the following:

MIL-PRF-39012

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2015-017)

Review activities:
Army – AT, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:11Z
Check the source to verify that this is the current version before use.